

Experiment 3

Installation and Basic Operation of Mininet

Mininet Topologies and Connectivity Testing

Student Name	Babin Bid
Roll Number	UG/02/BTCSE/2023/114
Registration Number	AU/2023/0009748
Section & Stream	Section B B.Tech CSE (Core)
Course & Semester	Cloud Computing Lab (CSE12045) · 7th Semester
Course Instructor	Prof. Anusuya Bera (Dept. of CSE)
Institution	ADAMAS University, Kolkata

1. Objective

Install and verify Mininet in the Ubuntu virtual machine, confirm the installed Mininet version, create simple built-in Mininet topologies, and test host-to-host connectivity using Mininet's pingall command.

2. Requirements

Ubuntu virtual machine with Mininet installed and working from the terminal.

3. Procedure

Step 1 — Download and Install Mininet in Ubuntu VirtualBox

Open the Ubuntu terminal inside the VirtualBox virtual machine and update the package information:

```
sudo apt update
```

Install Mininet directly from the Ubuntu package repository using:

```
sudo apt install mininet -y
```

After installation, verify that Mininet is installed correctly:

```
mn --version
```

The experiment screenshot shows Mininet version 2.3.0, confirming that Mininet is available in the Ubuntu VirtualBox environment.

Step 2 — Verify the Mininet Installation

Open the Ubuntu terminal and check the installed Mininet version using the following command:

```
mn --version
```

The captured terminal output shows Mininet version 2.3.0.

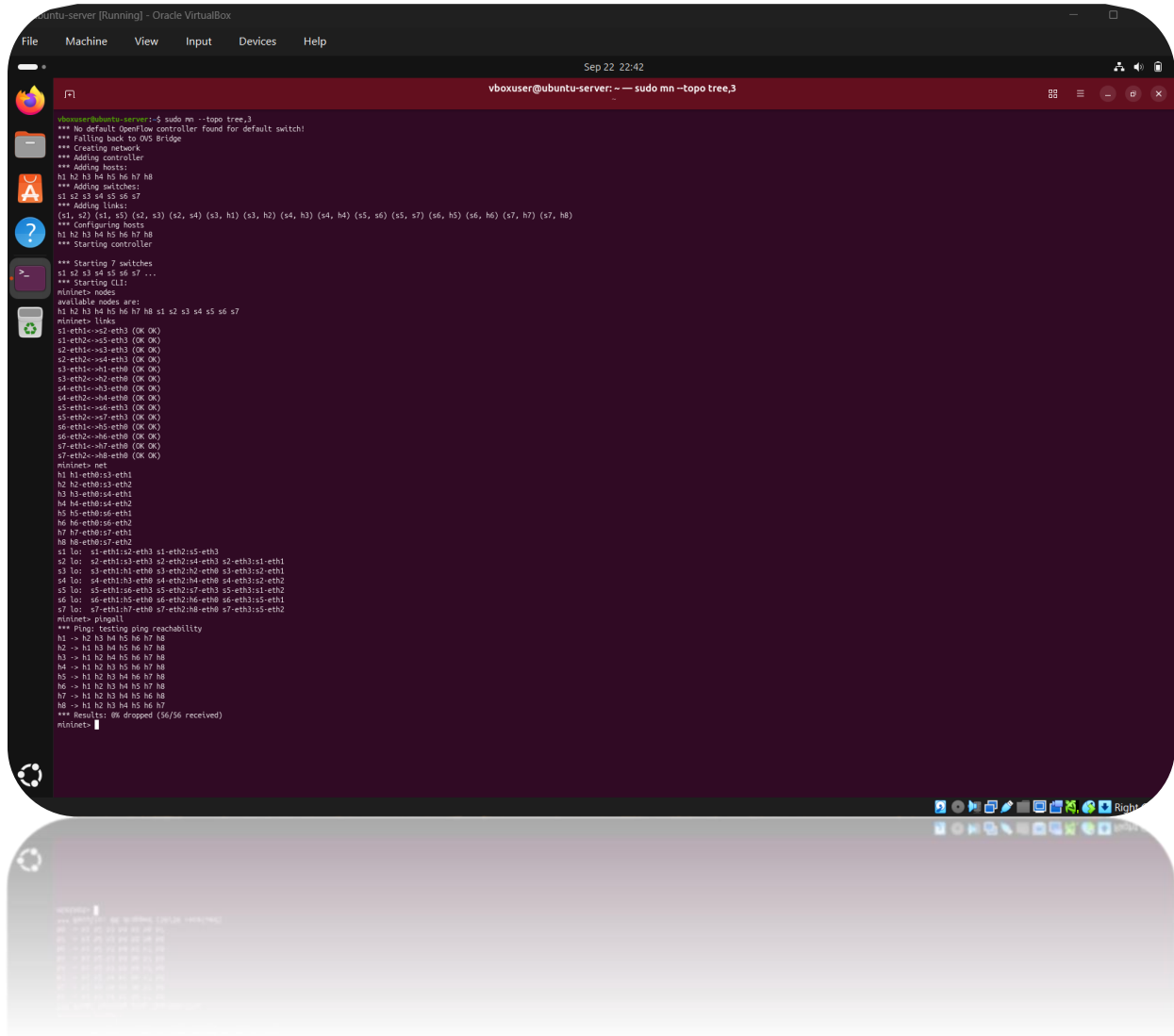


Figure 1 — Mininet version verification showing version 2.3.0.

Step 3 — Run the Mininet Built-in Ping Test

Mininet provides a built-in test that creates a small topology and checks whether the hosts can communicate. Execute:

```
sudo mn --test pingall
```

The screenshot shows two hosts, h1 and h2, connected through switch s1. The test reports 0% dropped (2/2 received), confirming successful connectivity.

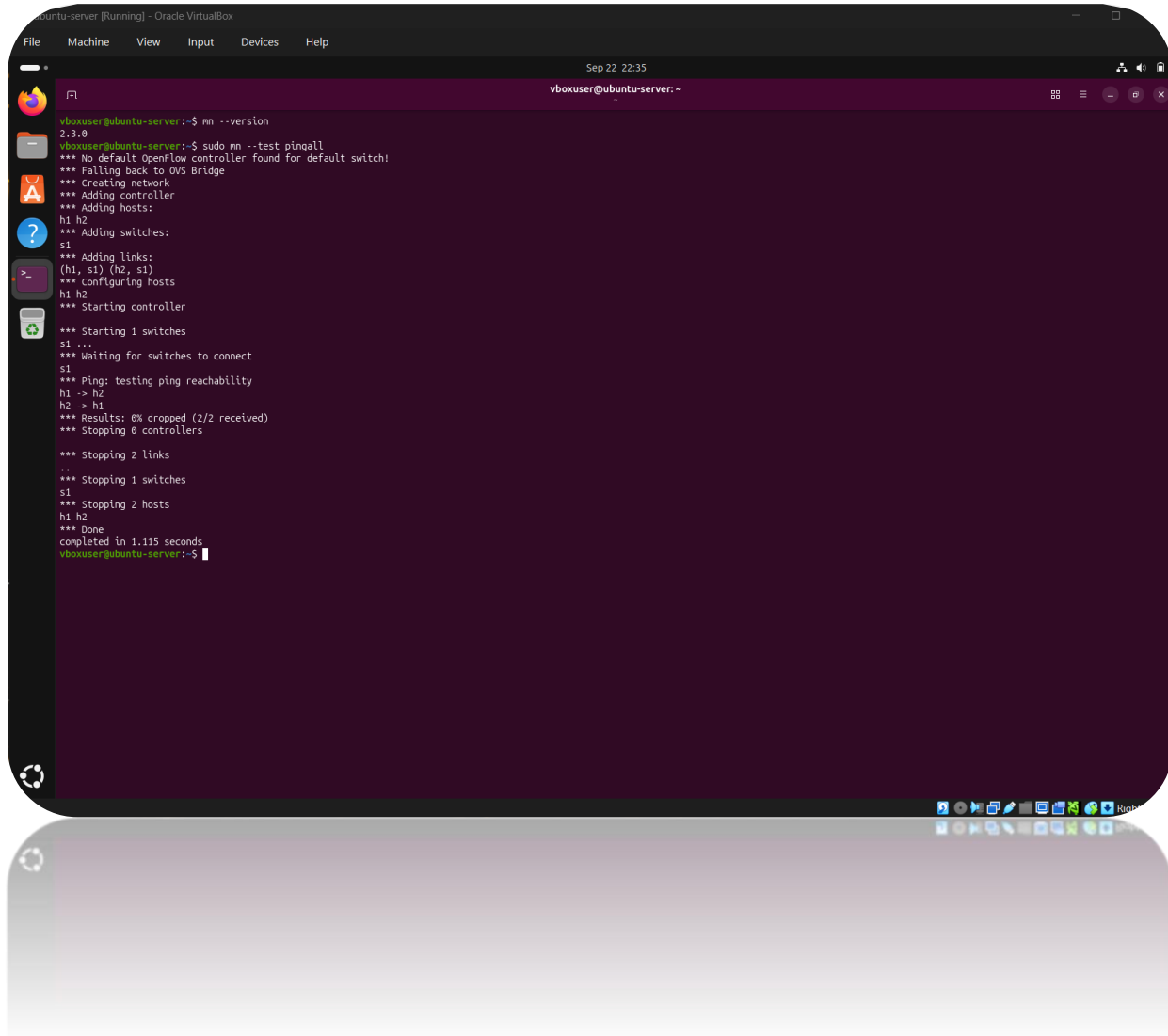


Figure 2 — Mininet pingall test with 0% packet loss.

Step 4 — Create a Single-Switch Topology with Three Hosts

Create a topology containing one switch and three hosts using the single topology option:

```
sudo mn --topo single,3
```

Mininet creates hosts h1, h2 and h3 and connects them to switch s1. At the Mininet CLI, the links can be inspected with:

```
mininet> links
```

The captured output shows the three host-to-switch links and the successful pingall result with 0% dropped (6/6 received).

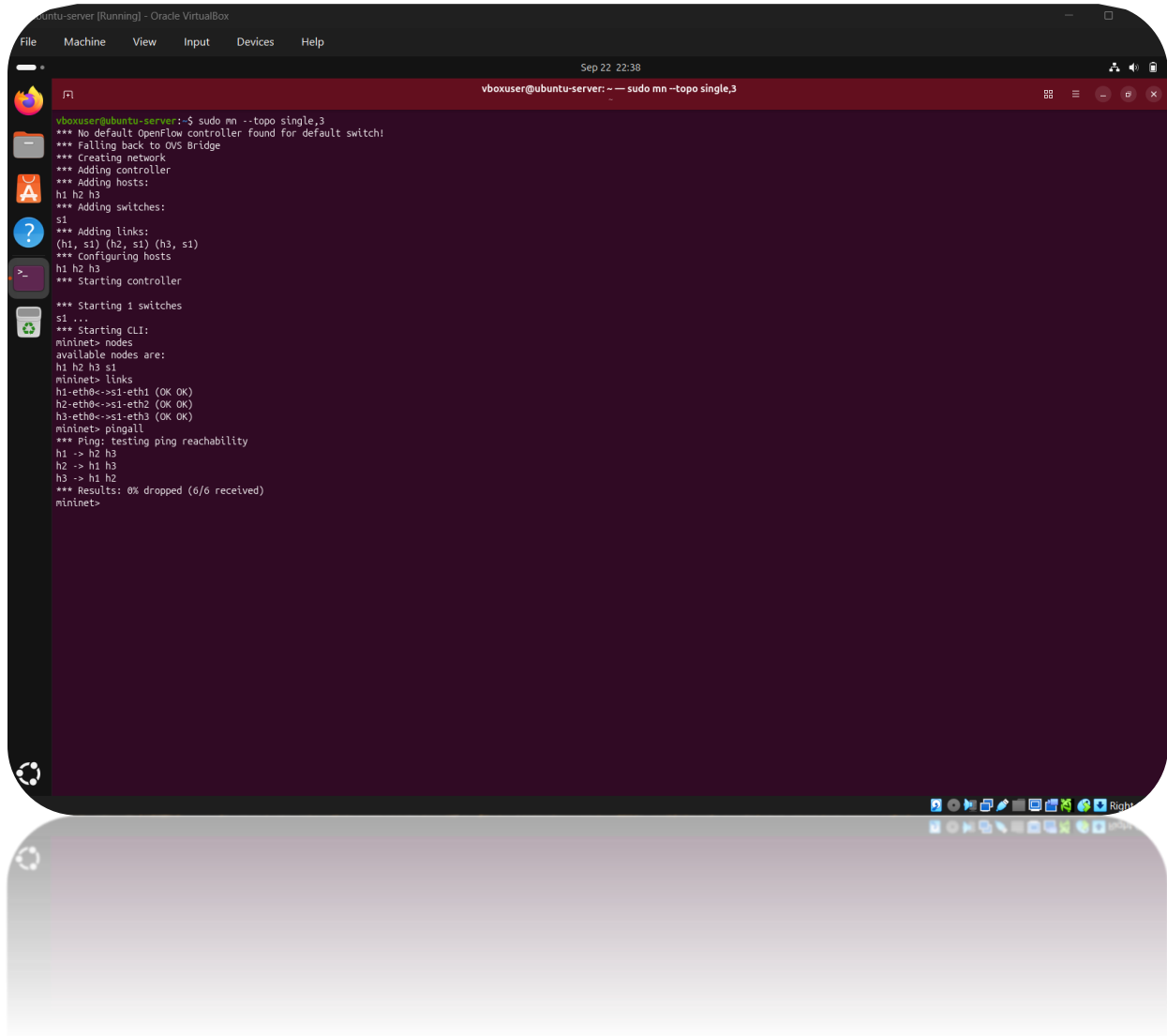


Figure 3 — Single-switch topology with three hosts and successful pingall connectivity.

Step 5 — Create a Tree Topology with Depth 3

Create a tree topology with depth 3 using:

```
sudo mn --topo tree,3
```

Mininet creates seven switches (s1-s7) and eight hosts (h1-h8). The terminal output lists the links between switches and hosts. Run the connectivity test from the Mininet CLI:

```
mininet> pingall
```

The captured result reports 0% dropped (56/56 received), showing successful reachability among all eight hosts in the generated tree topology.

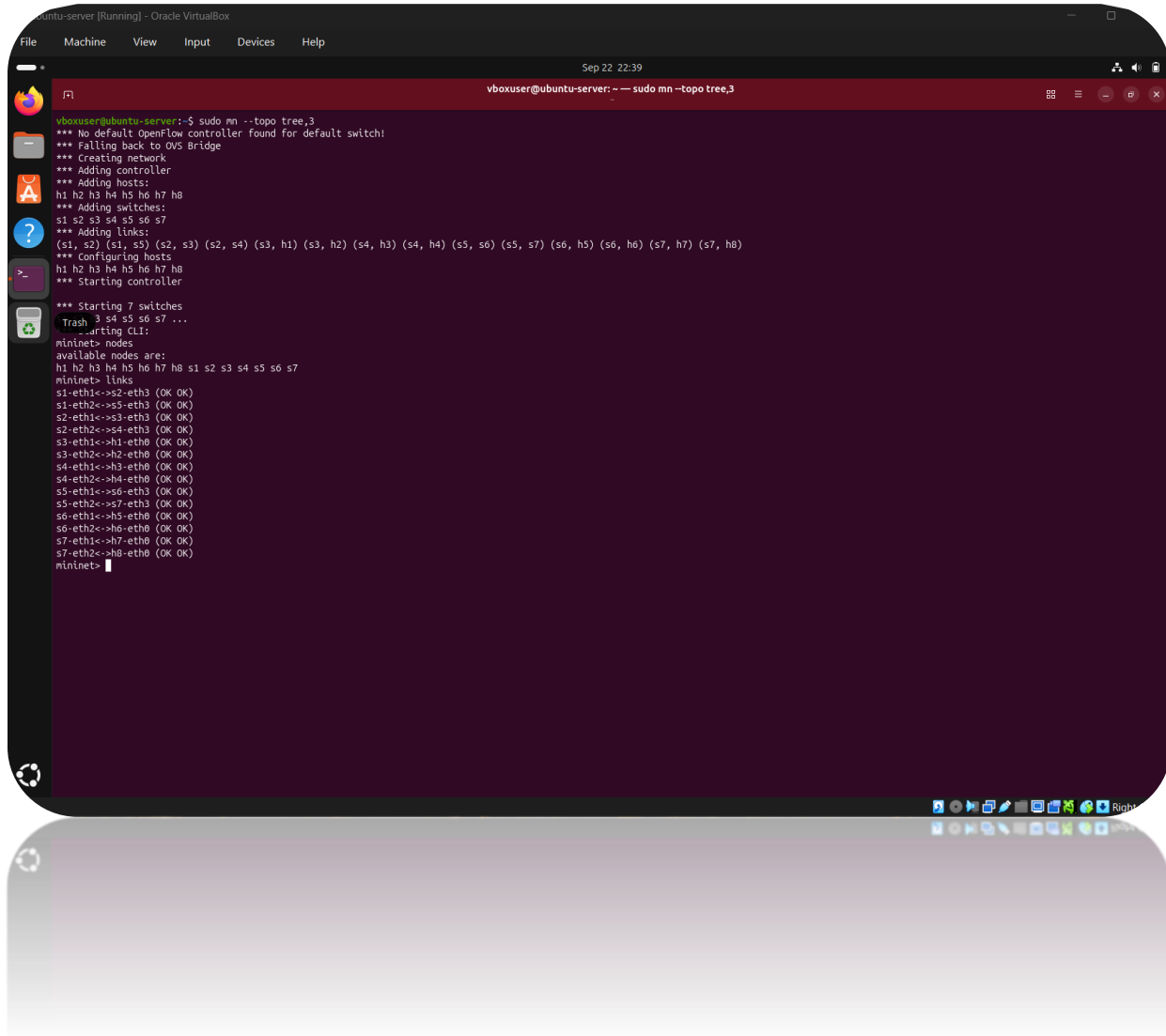


Figure 4 — Tree topology with depth 3, showing seven switches, eight hosts and successful pingall.

4. Important Mininet Commands Used

The following commands were used in the experiment:

```

mn --version
sudo mn --test pingall
sudo mn --topo single,3
sudo mn --topo tree,3
mininet> links
mininet> pingall

```

The sudo prefix is required for commands that create and manage the Mininet virtual network environment. Commands entered after the mininet> prompt are Mininet CLI commands.

5. Result

Mininet version 2.3.0 was verified successfully. The built-in ping test, a single-switch topology with three hosts, and a tree topology with depth 3 were executed successfully. The captured pingall results show 0% packet loss for the tested topologies.

6. Conclusion

The experiment demonstrated the basic operation of Mininet and its built-in topology creation facilities. Different virtual network structures were created and their host connectivity was verified using pingall.